

2024 2nd semester – CPIT220 MIDTERM – Male Section (23)

SOLVED

Question 1: Answer the following questions about the Introduction to IT
[8 Marks]

Q1.1: True or False [4 Marks]

True or False	T/F
Information Technology is systems of hardware and/or software that capture, process, exchange, store and/or present information using electrical, magnetic and/or electromagnetic energy	T
The Information Processing Cycle consists of two basic operations: input and output	F
Computer science or computing science (abbreviated CS) is the study of the theoretical foundations of information and computation, and of practical techniques for their implementation and application in computer systems	T
Systems engineering is concerned with all aspects of computer-based systems development including hardware, software and process engineering.	T

Q1.2: Fill in the following gaps with either 'information' or 'data' only? [2 Marks]

- **Data** is in a raw form.
- **Information** is the processed form of **data**.
- **Information** depends **data**.

Q1.3: List two application domains of Information Technology? [2 Marks]

1. **Education**
2. **Software Development**

Question 2: The following questions about Understanding the Digital Domain

[4 Marks]

Q2.1: Fill in the blanks

Noise	Digitization	Data rate	Encryption
Quantization	Encoding	Digital compression	Sampling

- The process of converting information to a binary form is called **digitization**
- Quantization** is the process of rounding off each sample to the closest level.
- Data rate** measured in terms of number of bits that can be sent per second over a transmission medium.
- Noise** is defined as unwanted occurrence that creates interference with a signal.
- Encryption** ensures that if third parties intercept information-carrying signal, they cannot decipher the signals.
- Digital Compression** is used to reduce the amount of stored data without losing the content.
- Sampling** is the process of recording an analog signal at regular discrete moments of time.
- Encoding** is the process of converting the human readable code into machine codes.

Question 3: The following questions about the Internet and the World Wide Web

[8 Marks]

Q3.1: Match the two columns.

[7]	Blog	1	Provides methods for packing and transmitting Information
[4]	Browser cache	2	Gathers data from a system without knowledge
[5]	Plug-ins	3	Programs that roam the web to add new web pages to search engine indexes
[2]	Spyware	4	Stores web page files on a computer when the user visits a site for the first time
[3]	Spiders	5	Additional software programs to extend the ability of the browser
[8]	HTML	6	Used to recognize information requests, process the requests, and send the requested documents
[1]	TCP/IP	7	Internet equivalent of a journal or diary where users post opinions and thoughts.
[6]	Web server	8	Use tags to specify how a web page should display

Question 4: Answer the following questions about System software

[4 Marks]

Q4.1: Which from the list is a basic function of Operating System [2 Marks]

- ☒ Starts the computer
- ☒ Manages applications
- ☒ Handles input and output device messages
- ☐ Shut down the computer
- ☐ Send Email before crashing
- ☒ Provides a user interface for communication

Q4.2: List two types of user interfaces [2 Marks]

1. Graphical user interface
2. Menu-driven user interface

Question 5: Answer the following questions about Inside the System Unit [6 Marks]

Q5.1: Define System Unit? List Three of its major components? [4 Marks]

A System Unit is: Case that contains the major hardware components of a computer

1. Motherboard
2. CPU
3. Power supply

Q5.2: Put the machine cycle below, in the right order? [2 Marks]

Decode	Fetch
Fetch	Decode
Store	Execute
Execute	Store